Multi-Task Dynamic Pricing in Credit Market with Contextual Information

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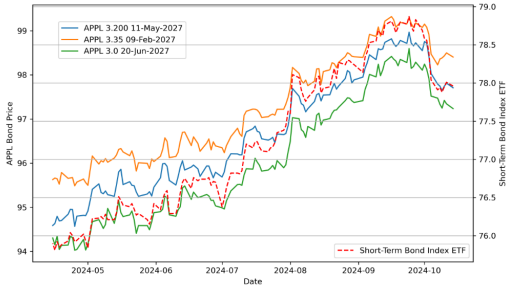
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Why interesting?

- **Need for real-time pricing in bond markets:**:
  - Real-time pricing is primarily studied in markets organized around a LOB.
  - however: bond: clients send request-for-quotes (RFQs), and dealers need to reply with prices immediately.
  - Most existing corporate bond pricing studies use low-frequency data(month)
- **Real-time pricing for every bond is difficult in practice:**
  - Historical data are limited because bonds trade infrequently： most liquid bonds only 300 times per day, while stocks in few minutes
  - Many bonds to price. There are about 66,000 U.S. corporate bonds, compared with about 4,500 stocks.
- Difficult: How to achieve real-time pricing in bond markets without a LOB?

Why interesting?

- Difficult: How to achieve real-time pricing in bond markets without a LOB?
- Bonds from the same firm or industry often share similar price patterns.



Three outstanding bonds (as of October 2024) issued by Apple and the Vanguard Short-Term Bond Index ETF. Apple bonds track each other and respond to the ETF in a similar fashion.

- This suggests using multi-task learning to price many bonds together, with shared common structure and bond-specific deviations.

Question

- Question: How can dealers price bonds in real time by using similarities across securities?
- tradition:
  - Individual learning: Each bond is priced separately, but data are limited.
  - Pooling: All bond data pooled together, but bond-level differences may ignored.
- multi-task pricing
  - Treat each bond as one task, and jointly learn a shared common component and a bond-specific idiosyncratic deviation.
  - result: outperforms the two strategies, or close to the best one.

Contribution

- credit bond markets pricing :low-frequency → real-time

	Monthly	Intra-Day	Real-Time
Commodities	Gorton et al. (2013), Gorton and Rouwenhorst (2006), Szymanowska et al. (2014)	Ready and Ready (2022)	LOB
FX	Lustig et al. (2011)	Andersen and Bollerslev (1998), Evans and Lyons (2002)	LOB
Equities	Fama (1970), Fama and French (1993)	Aleti and Bollerslev (2025), Aït-Sahalia et al. (2025)	LOB
Corporate bonds	Gebhardt et al. (2005), Chen et al. (2007)	(None)	**Our paper**

- Use multi-task learning for credit bond pricing.
  - prior: rely on full pooling or separate bond-level models.
  - extend : treats each bond as one task and shares information across similar bonds, helps address sparse trading data.

Design–multi-task learning

债券交易模型设定



t时刻请求购买债券j

发起询价



选择最低报价的做市商成交

我们报价p_t(收益率y_t)　最佳竞争对手报价$p_t^*(y_t^*)$

若$p_t \le p_t^*$,我们成交并获得收入p_t，否则收入为0

性能度量： $Regret(T) = \sum_{t=1}^{T}(reward(p_t^*) - reward(p_t))$

p_t^*: 用全样本ridge估计的参考报价benchmark

最佳竞争对手收益模型设定

$$y_{j,i}^* = \langle x_t, \theta_j^* \rangle + \varepsilon_{j,t}$$

x_t: 已观测的 feature

将每个证券的参数θ_j^*分解为：

$$\theta_j^* = \theta^* + \delta_j^*$$

所有债券共享的公共参数　债券j的个体偏差

债券相似度：　$\delta_{max} = \max_{j \in [M]} \|\delta_j^*\|_2$

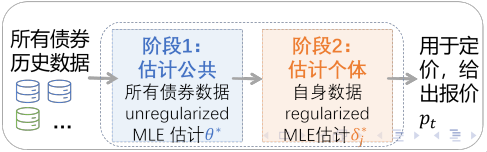
(越小，表示证券之间越相似)

Two stage multi-task pricing(TSMT)

核心思想：根据数据自适应地利用"共享结构"和"个体差异"。δ_{max}未知时，自适应权衡两种极端策略。

Pooling
$\delta_{max} \to 0$时最优
把所有债券数据合在一起，用一个公共模型

Individual
δ_{max}很大时最优
为每个债券单独建模，不使用其他债券数据



Design–multi-task learning

Stage I: aggregating data

$$\bar{\boldsymbol{\theta}}_{(k)} = \arg\min_{\boldsymbol{\theta}\in\mathbb{R}^d} \bar{\mathcal{L}}_{(k)}(\boldsymbol{\theta}), \quad \text{with } \bar{\mathcal{L}}_{(k)}(\boldsymbol{\theta}) \stackrel{\text{def}}{=} -\frac{1}{\tau_{k-1}} \sum_{t=\tau_{k-1}}^{\tau_k-1} \ell_t(\boldsymbol{\theta}).$$

Stage II: refine the estimation for every $j \in [M]$

$$\hat{\boldsymbol{\theta}}^j_{(k)} = \arg\min_{\boldsymbol{\theta}^j\in\mathbb{R}^d} \mathcal{L}^j_{(k)}(\boldsymbol{\theta}^j) + \lambda^j_{(k)} \left\|\boldsymbol{\theta}^j - \bar{\boldsymbol{\theta}}_{(k)}\right\|_2,$$

不要让这支债券偏离共同规律太远

$$\text{with} \quad \mathcal{L}^j_{(k)}(\boldsymbol{\theta}^j) \stackrel{\text{def}}{=} -\frac{1}{N^j_{(k)}} \sum_{t=\tau_{k-1}}^{\tau_k-1} \mathbb{1}[Z_t = j]\, \ell_t(\boldsymbol{\theta}^j),$$

d:特征维度
M:债券数量

$$\text{and} \quad \lambda^j_{(k)} = \sqrt{\frac{8u_F^2\bar{x}^2 d\log(2d^2M)}{N^j_{(k)}}}, \; N^j_{(k)} \stackrel{\text{def}}{=} \sum_{t=\tau_{k-1}}^{\tau_k-1} \mathbb{1}[Z_t = j].$$

债券j在上阶段出现次数少，应更依赖共同规律

For each time point t in the kth episode, set $\hat{\boldsymbol{\theta}}_t = \hat{\boldsymbol{\theta}}^{Z_t}_{(k)}$ and quote according to

$$p_t = p_t^\star\left(\langle \mathbf{x}_t, \text{Proj}_{\mathcal{B}(W)}(\hat{\boldsymbol{\theta}}_t)\rangle\right)$$

$$\lambda^j_{(k)} \propto \sqrt{\frac{\text{模型复杂度}}{\text{第 } j \text{ 支债券的数据量}}}$$

where $\text{Proj}_{\mathcal{B}(W)}$ is defined in (13) and $p_t^\star$ is defined in (9).

λj足够大，能够压住债券j由于样本少产生的随机噪声，这个量大致等于"单只债券参数估计误差的理论上界"

end for

Design: Synthetic Data and Validation

- **Data set:** Bonds share common structure but have bond-level diff:

$$\theta_j^\star = \theta^\star + \delta_j^\star, \quad j \in [M].$$

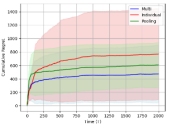
- **Features and regularization:**

$$x_t \sim N(0, I_d), \qquad \lambda_j(k) = 0.1\sqrt{\frac{d}{N_j(k)}}.$$

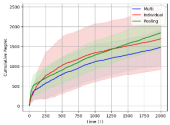
  - Fewer samples imply stronger regularization.

- **Strategies:** Pooling, Individual learning, and Multi-task learning.
- **Validation:** Compare accumulated regret by varying:
  - heterogeneity $\delta_{\max}$;
  - number of bonds M.

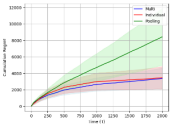
Result: Synthetic Data



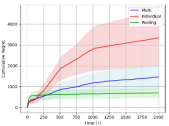
(a) $M = 2, \delta_{\max} = 0.1$.



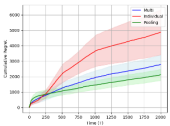
(b) $M = 2, \delta_{\max} = 0.5$.



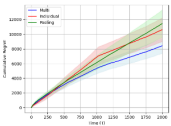
(c) $M = 2, \delta_{\max} = 2$.



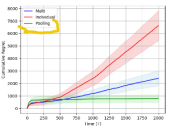
(d) $M = 10, \delta_{\max} = 0.1$.



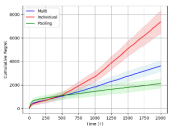
(e) $M = 10, \delta_{\max} = 0.5$.



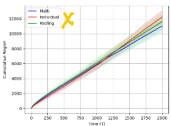
(f) $M = 10, \delta_{\max} = 2$.



(g) $M = 50, \delta_{\max} = 0.1$.



(h) $M = 50, \delta_{\max} = 0.5$.



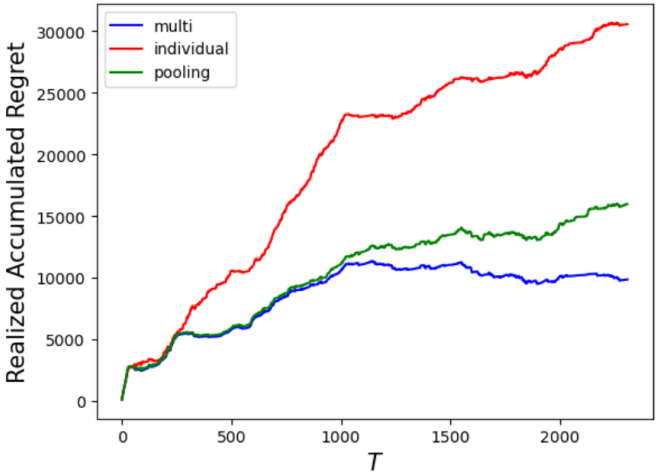
(i) $M = 50, \delta_{\max} = 2$.

- Larger M: pooling improves.
- Larger $\delta_{\max}$: individual learning improves.
- Multi-task learning is best or close to the best.

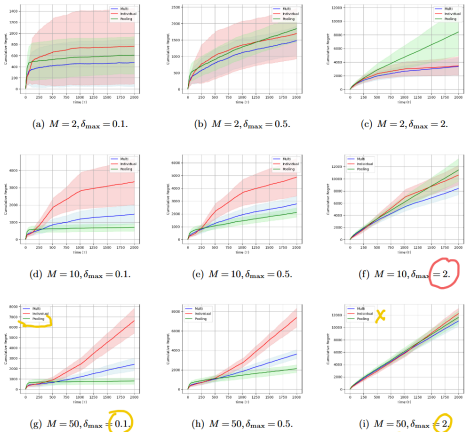
Design–real-world dataset on U.S. corporate bonds

- **Data:** FISD bond terms, LSEG bond features, and TRACE transactions.
- **Sample:** 37 high-tech and energy bonds from the 500 most traded bonds during 2023/09/18–2023/09/29; only dealer-side sell trades are used.
- **Features X:** 4 PCs from bond features, plus 30-trade rolling average price, and rolling volume.
- **Oracle benchmark:** Full-sample ridge reg is used to estimate oracle prices.
  - The final sample keeps 30 bonds with $R^2 > 40\%$.
- **Evaluation:** Accumulated regret of pooling, individual, and multi-task.

Result–real-world dataset on U.S. corporate bonds



复现：主要结果 1--模拟数据

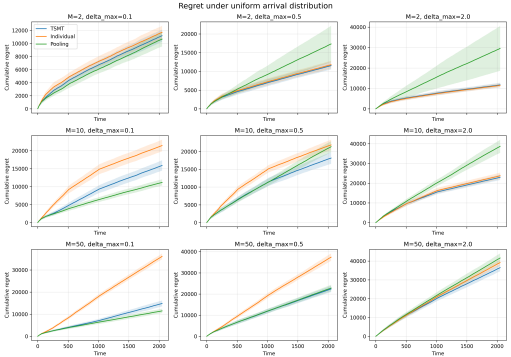


(a) $M = 2, \delta_{\max} = 0.1$. (b) $M = 2, \delta_{\max} = 0.5$. (c) $M = 2, \delta_{\max} = 2$.

(d) $M = 10, \delta_{\max} = 0.1$. (e) $M = 10, \delta_{\max} = 0.5$. (f) $M = 10, \delta_{\max} = 2$.

(g) $M = 50, \delta_{\max} = 0.1$. (h) $M = 50, \delta_{\max} = 0.5$. (i) $M = 50, \delta_{\max} = 2$.

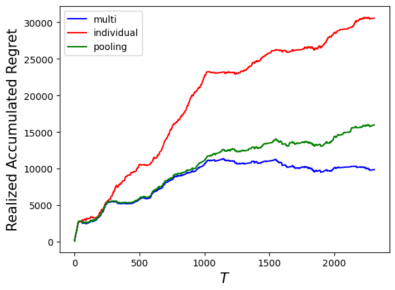
图 1: 原文结果



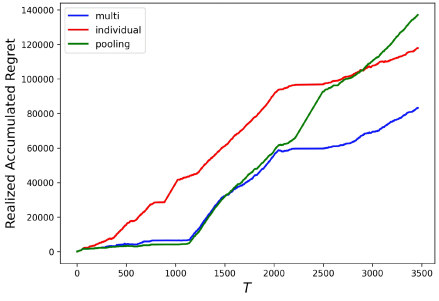
- M 增加，pooling; 异质性增加，indi
- muti 最优（5、6）和接近最优

复现：主要结果 2--现实数据

原文结果



复现结果



- 符合原文结论：muti 累计 regret 逐渐呈现优势，优于其他策略（22；34；40）
- 差异来源 (虽然结论一致，但数值、趋势、时间长度有区别)：
  - 交易样本数量，本文按原文描述筛选后得到约 3500 个交易，原文 2000 个；
  - 特征构造，原文仅列举部分特征 +others，我使用文中明确的 5 个特征提取 PC4;
  - 特征数据来源，原文使用 LSEG Workspace，本文使用 WRDS Dickerson 数据。

Thanks!

Question & idea?

Idea

- 中国市场应用：
  - 中国公司债市场更加面临交易稀疏问题，不少债券一个月都没有交易。但直接考察高频交易不好讲故事？
  - 债市的收益更新慢导致债基风险累计，可以从日度或者月度的“实时”定价及基金市场风险预测来做。
- 方法优化：
  - 在选择相似债券的方面很关键，选用同行、市场指数、相似特征的债券
  - 上周郭康的那篇也是提取横截面不同股票之间的信息
  - 股债之间也可以使用这种 muti 任务思路？